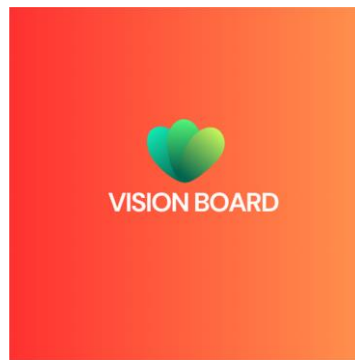




# AZURE DATA ENGINEERING 4 MONTHS COURSE, DP900 DAY-1



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# WHAT IS CLOUD COMPUTING?

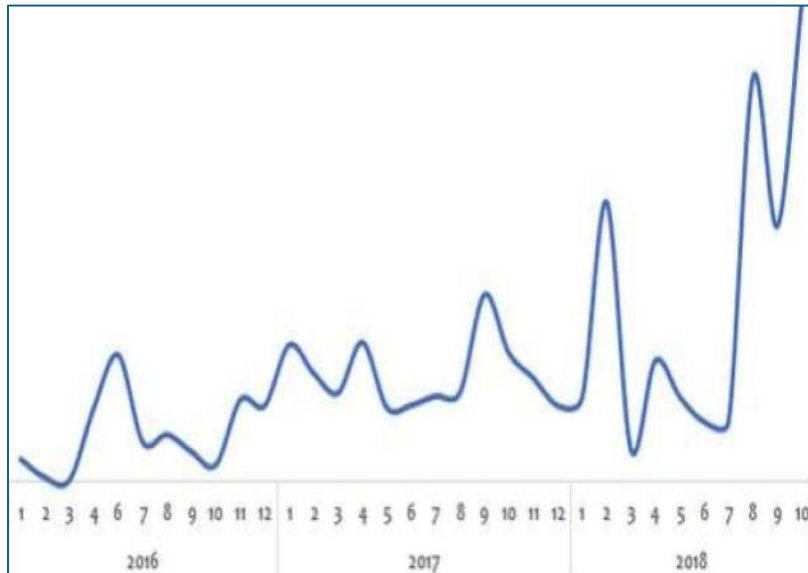


- Cloud Computing means storing and accessing the data and programs on remote servers that are hosted on the internet instead of the computer's hard drive or local server. The data that is stored can be files, images, documents, or any other storable document.
- Cloud computing helps users in easily accessing computing resources like storage, and processing over internet rather than local hardware.

Major Features Include:

- **On-Demand Access:** Users can access cloud services and resources based on-demand they can scale up or down the without having to invest for physical hardware.
- **Types of Services:** Cloud computing offers various benefits such as cost saving, scalability, reliability and accessibility it reduces capital expenditures, improves efficiency.

# Example: Online Shopping Application



- Challenges:
  - Higher usage during Holidays and weekends
  - Lesser loads during rest of the time
  - Usage Peaks during sales and offer time
- Companies need to adjust infrastructure according to the user demands – higher expense
- By moving to cloud they can procure more resources only when necessary and pay for the resources actually used – lesser capital cost, lower expense

# Let's compare.

## Before Cloud



- High cost of procuring infrastructure
- Needs ahead-of-time planning
- Low infrastructure utilization during off peak hours
- Dedicated infrastructure maintenance team

## After cloud



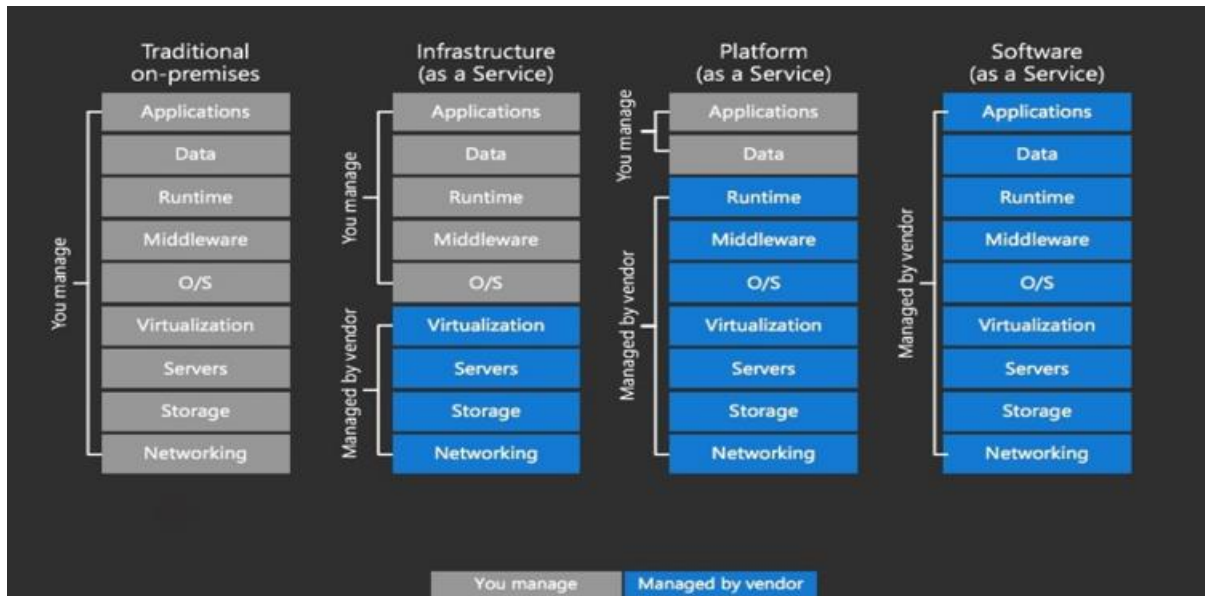
- Infrastructure set up by cloud providers
- Can provision resources as required
- resources can be expanded during peak hours
- No maintenance team required, support provided by cloud providers

## AZURE OVERVIEW

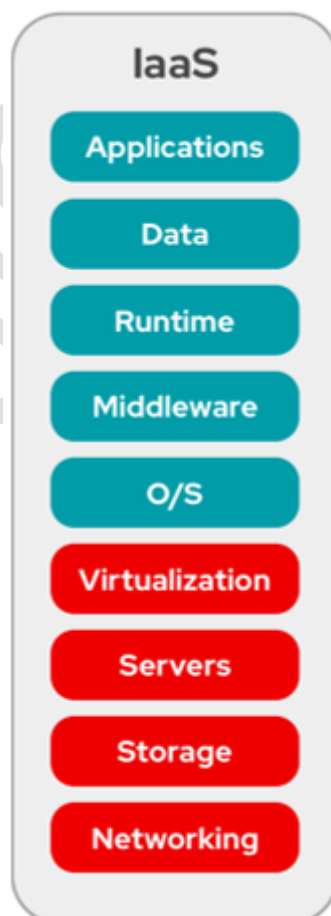


- Azure is a cloud computing platform provided by Microsoft. It offers a wide range of services, including computing, storage, networking, databases, analytics, and more.
- Azure enables businesses to build, deploy, and manage applications and services through Microsoft's global network of data centres.
- **Scalability:** Azure provides elastic scaling, enabling businesses to expand resources as needed without significant upfront investment.
- **Flexibility:** supports a wide range of workloads, from simple web applications to complex data analytics, catering to diverse business requirements.
- **Cost Efficiency:** With pay-as-you-go pricing and cost optimization tools, Azure helps businesses minimize expenses while maximizing performance.
- **Security and Compliance:** Azure implements robust security measures and compliance certifications, ensuring data protection and regulatory adherence.

# Types of Azure Services

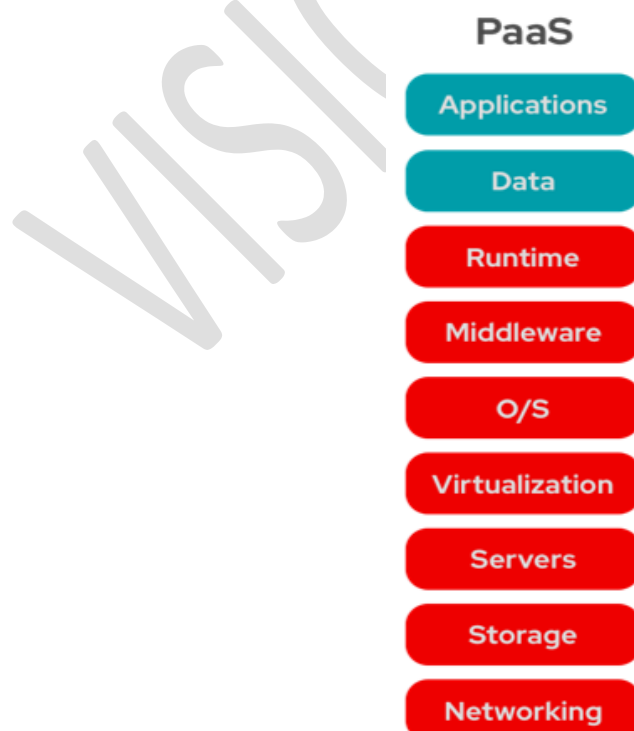


## INFRASTRUCTURE AS A SERVICE (IAAS)



- Virtual machines, storage, and networking will come under the category of infrastructure as a service but the users have to do manually the build and deploy of the applications. Azure will support a wide range of operating systems because of its Hyper-hypervisor.
- Only infrastructure is provided.
- **Cloud Provider is responsible for:**
  - Virtualization, Hardware and Networking
- **You are responsible for:**
  - OS upgrades and patches
  - Database software and upgrades
  - Database Configuration (Tables, Indexes, Views etc) Data Scaling of compute & storage, Availability and Durability

## PLATFORM AS A SERVICE (PAAS)



- Azure app service, Azure functions, and logic apps are some services that are offered by Azure under the platform as a service. This service will provide autoscaling and load balancing and also there will be a pre-configured environment for the application
- **Cloud provider is responsible for:**
  - Virtualization, Hardware and Networking
  - OS upgrades and patches Database software and upgrades Scaling, Availability, Durability etc.
- **You are responsible for:**
  - Database Configuration (Tables, Views, Indexes)
  - Data
- **Examples:** Azure SQL Database, Azure Cosmos DB and a lot more.
- You will **NOT** have access to OS and Database software (most of the times!)

## SOFTWARE AS A SERVICE (SAAS)



- Office 365, Dynamics 365, and Azure Active Directory are some of the services provided by Microsoft Azure under Software as a Service (SaaS) the complete application will be managed by the Microsoft azure including deploying, scaling and load balancing
- Offered on a subscription basis (pay-as-you-go)
- Customer relationship management (CRM), enterprise resource planning (ERP) and do tools
- **Cloud provider is responsible for:**
  - OS (includes upgrades and patches) Application Runtime
  - Auto scaling, Availability & Load balancing etc.
  - Application code and/or Application Configuration
  - Configuring memory and instances
- **Customer is responsible for:**
  - Configuring the software!

## AZURE DATA STORAGE INFRASTRUCTURE

- Azure's data storage infrastructure forms the backbone of its cloud services, providing scalable and reliable storage solutions for businesses of all sizes. It encompasses a diverse range of storage options, including Blob Storage, File Storage, Table Storage, Queue Storage, and Disk Storage, catering to various data management needs.
- Azure's data storage infrastructure is designed to handle vast amounts of data, from small files to massive datasets, with high performance and low latency. **The infrastructure is distributed across Azure's global network of data centres(regions), ensuring data proximity and availability for users around the world.**

- Cloud computing relies heavily on data storage infrastructure for storing application data, files, databases, backups, and other critical assets securely in the cloud. Azure's data storage services enable businesses to scale their storage resources dynamically based on demand, reducing upfront costs and operational complexity.
- Data stored in Azure's infrastructure benefits from built-in redundancy, encryption, compliance, and disaster recovery capabilities, enhancing data protection and regulatory compliance. In today's data-driven world, Azure's data storage infrastructure plays a crucial role in facilitating digital transformation, innovation, and competitive advantage for businesses across industries.

## Azure Regions

### Scenario 1 :

- Imagine that your application is deployed in a data centre in US. **What would be the challenges?**
- Challenge 1 : Slow access for users from other parts of the world (high latency)
- Challenge 2 : What if the data centre crashes? Your application goes down (low availability)



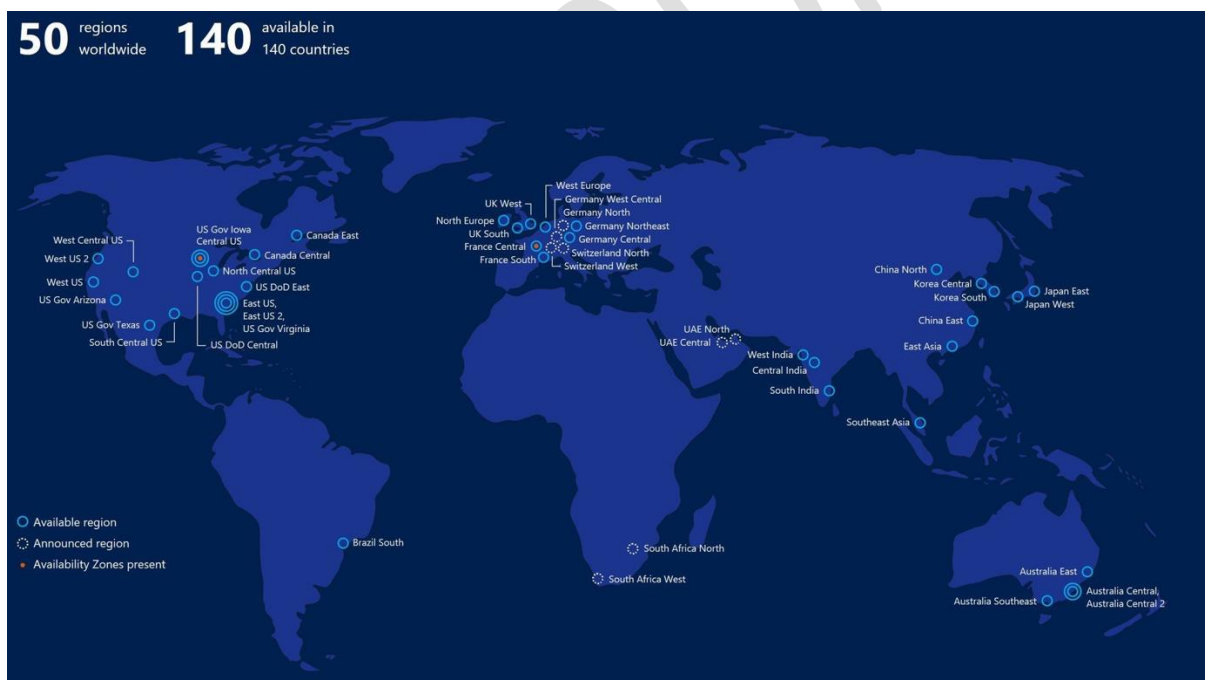
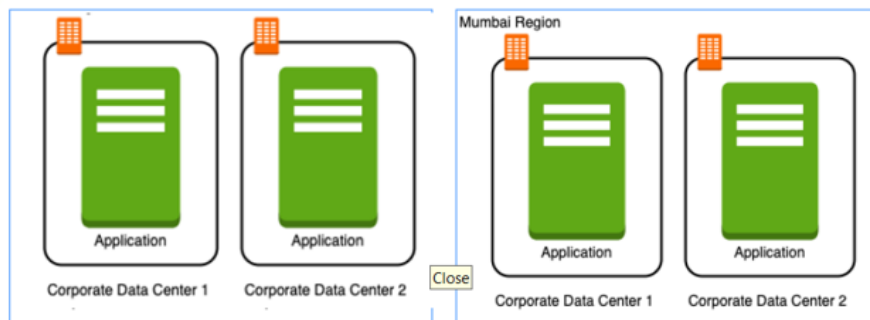
### Scenario 2:

- Let's add in one more data centre in US. What would be the challenges?
- Challenge 1 : Slow access for users from other parts of the world
- Challenge 2 .What if one data centre crashes?  
- this issue is now solved as your application is still available from the other data centre
- Challenge 3 : What if entire region of US is unavailable?  
- Your application goes down



### Scenario 3 :

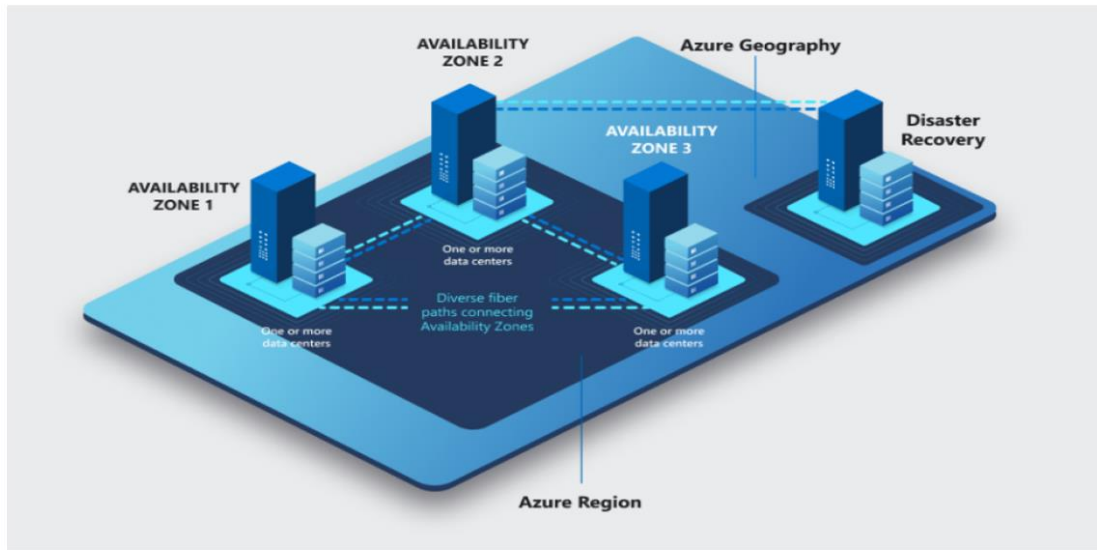
- Let's add a new region : Mumbai, What would be the challenges?
- Challenge 1: slow access for users from other parts of the world. This can be partly solved adding deployments for your applications in other regions.
- Challenge 2: What if one data centre crashes. This is solved as your application is still live from the other data centres
- Challenge 3 : What if entire region of US is unavailable? This is solved as Your application is served from Mumbai



- Azure regions are geographically distributed data centre locations where Azure resources and services are hosted, providing customers with options to deploy applications and store data closer to their users.

- Significance: Azure regions offer high availability, scalability, and data residency compliance, allowing businesses to optimize performance, meet regulatory requirements, and enhance user experience.
- Each region is a collection of data centers strategically located within a specific geographic area, ensuring redundancy and fault tolerance for Azure services.
- Azure regions are designed to provide low-latency connectivity, enabling fast and reliable access to cloud resources for users and applications worldwide.
- Businesses can leverage Azure regions to build resilient, globally distributed architectures, ensuring their applications remain available and responsive even in the face of disruptions or outages.
- **Advantages:**
  - High Availability
  - Low Latency Global Footprint
  - Adhere to government regulations

# Availability Zone



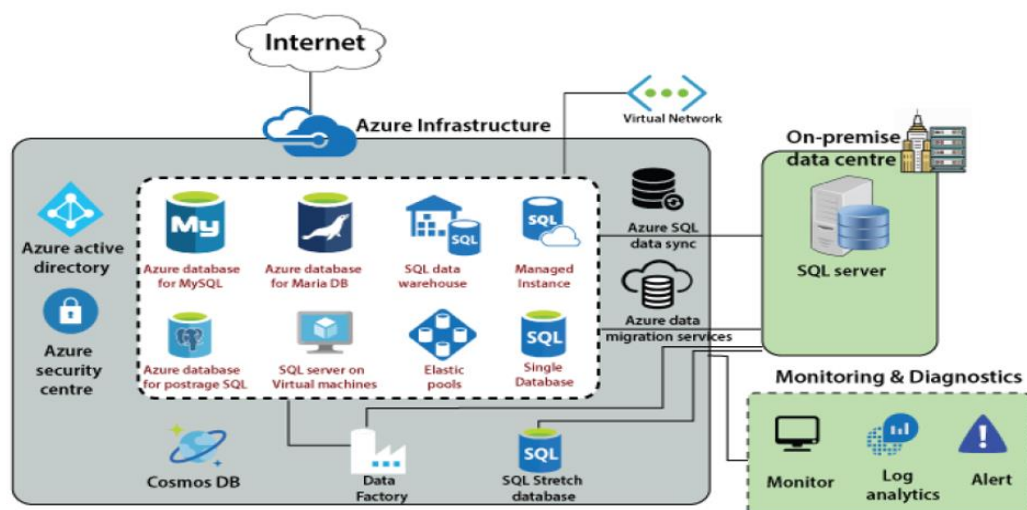
- Availability Zones are distinct physical locations within an Azure region, each comprising one or more data centres equipped with independent power, cooling, and networking infrastructure.
- These Zones are strategically positioned to mitigate the risk of localized failures, providing redundancy and fault tolerance for high availability applications and services. By distributing infrastructure across Availability Zones, Azure ensures that applications remain resilient to failures or disruptions within a single zone, enhancing overall service reliability.
- Each Availability Zone within an Azure region operates autonomously, minimizing the likelihood of simultaneous failures and ensuring continuous operation even in adverse conditions. Availability Zones enable customers to architect highly available, fault-tolerant solutions on Azure, offering peace of mind and resilience for mission-critical workloads and applications.

Note : **Not** all Azure regions have Availability Zones

| Regions                       | Availability Zones |
|-------------------------------|--------------------|
| (US) East US                  | 3                  |
| (US) West Central US          | 0                  |
| (Europe) West Europe          | 3                  |
| (Asia Pacific) Southeast Asia | 3                  |

Example

## Azure Data Services



# Data Formats

## Structured Data

- Data organized into a predefined format, often stored in relational databases.
- Examples: Tables in a SQL database, Excel spreadsheets, CSV files.
- Characteristics: Clearly defined schema, organized into rows and columns, suitable for querying and analysis.

## Semi-structured Data

- Data that has some structure but does not fit neatly into relational database tables.
- Examples: JSON files, XML files, log files.
- Characteristics: Contains tags, keys, or attributes that provide some structure, but data elements may vary in structure and format.

## Unstructured Data

- Data organized into a NOT A FIXED format, often stored in NON relational databases.

| Customer |           |          |                     |             |
|----------|-----------|----------|---------------------|-------------|
| ID       | FirstName | LastName | Email               | Address     |
| 1        | Joe       | Jones    | joe@litware.com     | 1 Main St.  |
| 2        | Samir     | Nadoy    | samir@northwind.com | 123 Elm Pl. |

| Product |             |       |
|---------|-------------|-------|
| ID      | Name        | Price |
| 123     | Hammer      | 2.99  |
| 162     | Screwdriver | 3.49  |
| 201     | Wrench      | 4.25  |

Structured data

```
JSON
// Customer 1
{
  "firstName": "Joe",
  "lastName": "Jones",
  "address": {
    "streetAddress": "1 Main St.",
    "city": "New York",
    "state": "NY",
    "postalCode": "10099"
  },
  "contact": [
    {
      "type": "home",
      "number": "555 123-1234"
    },
    {
      "type": "email",
      "address": "joe@litware.com"
    }
  ]
}
```

Semi-structured data



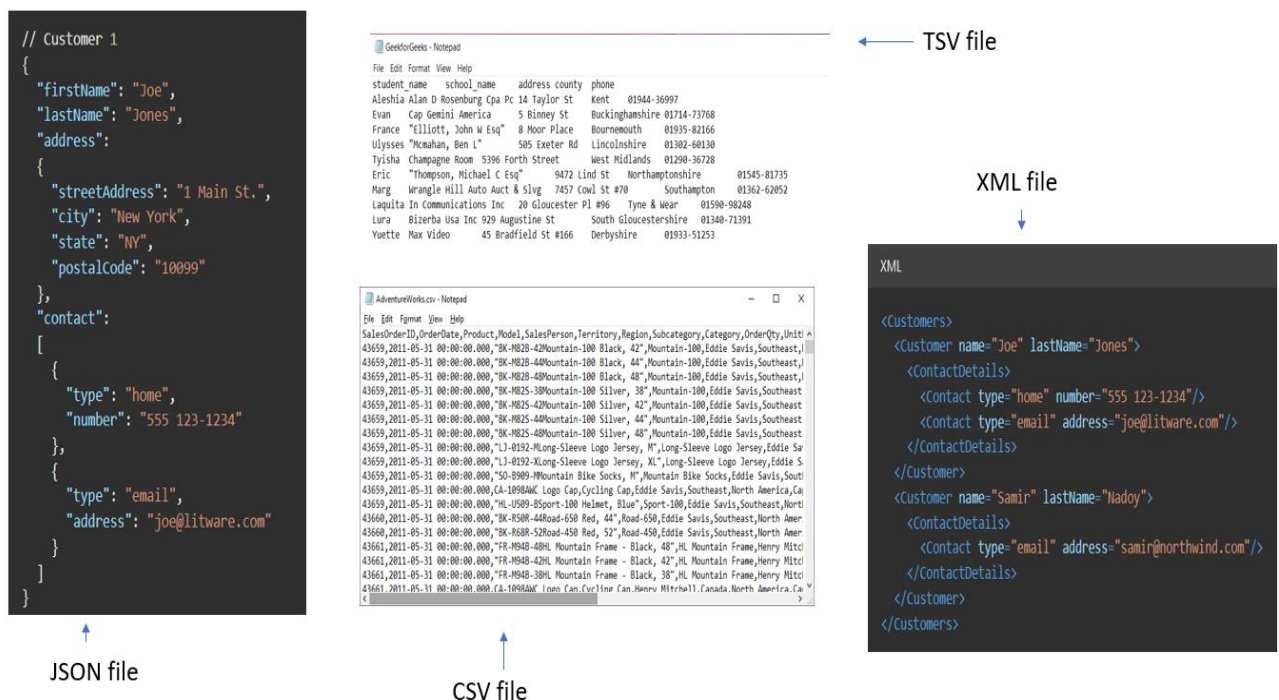
unstructured data

## FILE STORAGE

- How is the data stored? – it can be stored as files or in the database. Let us explore the file formats available to us.
- **File Formats:**
  - **Delimited text** - Delimited text formats like CSV, TSV, and space-delimited are commonly used for storing structured data, with fields separated by specific delimiters and rows terminated by carriage returns. These formats are human-readable and widely supported, making them suitable for

various applications and services requiring structured data access.

- **JavaScript Object Notation (JSON)** - lightweight, human-readable data format used for transmitting structured data between servers and web applications, based on JavaScript syntax. Its simplicity, readability, and broad support across platforms and programming languages make it a popular choice for data exchange in web services and modern application development.
- **Extensible Markup Language (XML)** - versatile and human-readable data format commonly used for structuring and storing data. It uses tags enclosed in angle brackets (< >) to define elements and attributes, allowing users to create customized document schemas tailored to their specific needs.



- **Binary Large Object (BLOB)**: All files are ultimately stored as binary data (1's and 0's). Some file formats, particularly for unstructured data, store data as raw binary. Common types of

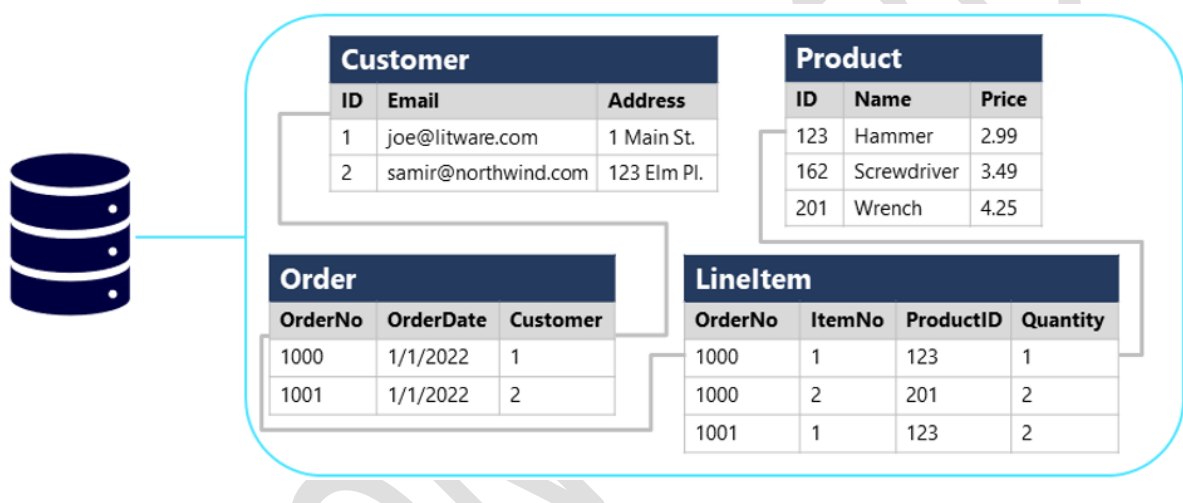
data stored as binary include images, video, audio, and application-specific documents. When working with data like this, data professionals often refer to the data files as BLOBs (Binary Large Objects).

- **Optimized file formats:**

- **Avro** is a row-based format. It was created by Apache. Each record contains a header that describes the structure of the data in the record. This header is stored as JSON. The data is stored as binary information. An application uses the information in the header to parse the binary data and extract the fields it contains. Avro is a good format for compressing data and minimizing storage and network bandwidth requirements.
- **ORC (Optimized Row Columnar format)** organizes data into columns rather than rows. It was developed for optimizing read and write operations in Apache Hive (Hive is a data warehouse system that supports fast data summarization and querying over large datasets). An ORC file contains stripes of data. Each stripe holds the data for a column or set of columns. A stripe contains an index into the rows in the stripe, the data for each row, and a footer that holds statistical information (count, sum, max, min, and so on) for each column.
- **Parquet** : Apache Parquet is a file format designed to support fast data processing for complex data, with several notable characteristics. It is column oriented, open source (free to use) and self-describing as the parquet file contains metadata including schema and structure.

# Databases:

- **Relational Database** : store structured data in tables, each with a unique primary key. This key is used for referencing data across tables; for instance, a customer's key can link to their sales orders. This normalization reduces data redundancy, with SQL used for managing and querying these tables consistently across different database systems.



- **Non relational Databases:** Non-relational databases are data management systems that don't apply a relational schema to the data. Non-relational databases are often referred to as NoSQL database, even though some support a variant of the SQL language

Key value based

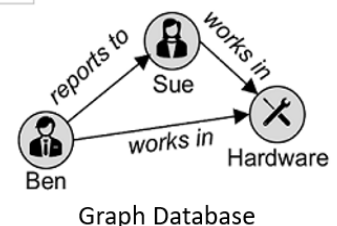
| Products |                        |
|----------|------------------------|
| Key      | Value                  |
| 123      | "Hammer (\$2.99)"      |
| 162      | "Screwdriver (\$3.49)" |
| 201      | "Wrench (\$4.25)"      |

Document databases

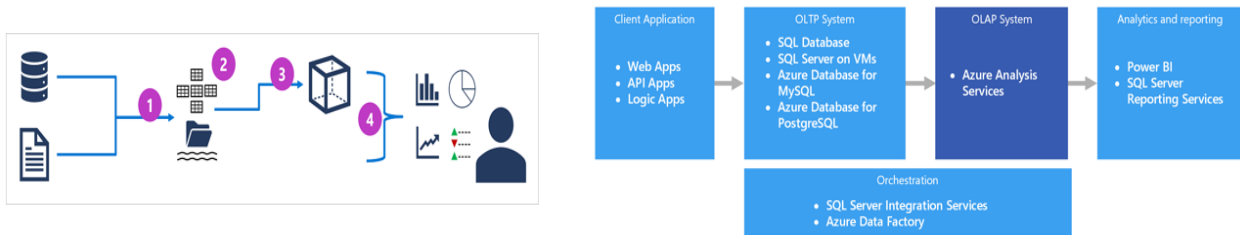
| Customers |                                                                    |
|-----------|--------------------------------------------------------------------|
| Key       | Document                                                           |
| 1         | {<br>"name": "Joe Jones",<br>"email": "joe@litware.com"<br>}       |
| 2         | {<br>"name": "Samir Nadoy",<br>"email": "Samir@northwind.com"<br>} |

Column family database

| Orders |             |             |         |       |
|--------|-------------|-------------|---------|-------|
| Key    | Customer    |             | Product |       |
|        | Name        | Address     | Name    | Price |
| 1000   | Joe Jones   | 1 Main St.  | Hammer  | 2.99  |
| 1001   | Samir Nadoy | 123 Elm Pl. | Wrench  | 4.25  |



# Analytical Data Processing



- Operational data is extracted, transformed, and loaded (ETL) into a data lake for analysis.
- Data is loaded into a schema of tables - typically in a Spark-based **data Lakehouse** with tabular abstractions over files in the data lake, or a data warehouse with a fully relational SQL engine → **OLTP**
- Data in the **data warehouse** may be aggregated and loaded into an online analytical processing (**OLAP**) model, or cube. Aggregated numeric values (measures) from fact tables are calculated for intersections of dimensions from dimension tables. For example, sales revenue might be totalled by date, customer, and product.
- The data in the data lake, data warehouse, and analytical model can be queried to produce reports, visualizations, and dashboards.
- **Data Warehouses:**
  - Established method for storing data in a relational schema optimized for read operations.
  - Primarily used for queries supporting reporting and data visualization.

- **Data Lake houses:**
  - Innovative approach combining the flexible storage of a data lake with relational querying semantics of a data warehouse.
  - Table schema may require denormalization for faster query performance, introducing some data duplication from OLTP sources.
- **OLTP(online transaction processing):**
  - The databases that a business uses to store all its transactions and records are called online transaction processing (OLTP) databases.
  - Databases that support heavy write operations, these are not optimised for analysis
  - Ensures ACID ( atomicity, consistency, isolation, Durability)
- **OLAP (Online Analytical Processing):**
  - Designed to help extract business intelligence information from the data in a highly performant way.
  - Optimised for heavy read operations
  - Aggregated data storage optimized for analytical workloads, Aggregations across dimensions at various levels allow drill up/down to view summaries hierarchically.

## User Roles in Data Analytical Workflows

- **Data Scientists:**
  - Work directly with data files in the data lake to explore and model data.
- **Data Analysts:**
  - Query tables in the data warehouse to produce complex reports and visualizations.
- **Business Users:**
  - Consume pre-aggregated data in analytical models through reports or dashboards for decision-making